1 Tangent Map

- Let

$$S_1 : \mathbf{r}_1 = \mathbf{r}_1(u_1, v_1), \quad (u_1, v_1) \in D_1 \subseteq \mathbb{R}^2,$$
$$S_2 : \mathbf{r}_2 = \mathbf{r}_2(u_2, v_2), \quad (u_2, v_2) \in D_2 \subseteq \mathbb{R}^2,$$

be two regular parametric surfaces. For any smooth map

$$\varphi : D_1 \longrightarrow D_2$$
$$(u_1, v_1) \mapsto (u_2, v_2) = \big(u_2(u_1, v_1), v_2(u_1, v_1)\big),$$

it induces a map between S_1 and S_2

$$\widetilde{\varphi} : S_1 \longrightarrow S_2$$
$$\mathbf{r}_1(u_1, v_1) \mapsto \mathbf{r}_2\big(u_2(u_1, v_1), v_2(u_1, v_1)\big),$$

which will be still denoted as $\varphi : S_1 \longrightarrow S_2$.

- Let $\varphi : S_1 \longrightarrow S_2$ be a smooth map defined as above, if there is smooth curve γ on S_1

$$\gamma : (t_0 - \varepsilon, t_0 + \varepsilon) \longrightarrow S_1$$
$$t \mapsto \gamma(t) = \mathbf{r}_1\big(u_1(t), v_1(t)\big),$$

then it induces another curve $\varphi \circ \gamma$ on S_2

$$\varphi \circ \gamma : (t_0 - \varepsilon, t_0 + \varepsilon) \longrightarrow S_2$$
$$t \mapsto \varphi\big(\gamma(t)\big) = \mathbf{r}_2\Big(u_2\big(u_1(t), v_1(t)\big), v_2\big(u_1(t), v_1(t)\big)\Big),$$

hence we see that the tangent vector for the curve $\varphi \circ \gamma$ on S_2 is

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}(\varphi \circ \gamma)(t) &= \frac{\partial \mathbf{r}_2}{\partial u_2} \cdot \frac{\mathrm{d}u_2(t)}{\mathrm{d}t} + \frac{\partial \mathbf{r}_2}{\partial v_2} \cdot \frac{\mathrm{d}v_2(t)}{\mathrm{d}t} \\ &= \frac{\partial \mathbf{r}_2}{\partial u_2} \cdot \left(\frac{\partial u_2}{\partial u_1} \cdot \frac{\mathrm{d}u_1(t)}{\mathrm{d}t} + \frac{\partial u_2}{\partial v_1} \cdot \frac{\mathrm{d}v_1(t)}{\mathrm{d}t}\right) + \frac{\partial \mathbf{r}_2}{\partial v_2} \cdot \left(\frac{\partial v_2}{\partial u_1} \cdot \frac{\mathrm{d}u_1(t)}{\mathrm{d}t} + \frac{\partial v_2}{\partial v_1} \cdot \frac{\mathrm{d}v_1(t)}{\mathrm{d}t}\right).\end{aligned}$$

Definition 1.(Tangent Map)
Let $\varphi : S_1 \longrightarrow S_2,\ (u_1, v_1) \mapsto \varphi(u_1, v_1) = (u_2, v_2) = \big(u_2(u_1, v_1), v_2(u_1, v_1)\big)$ be a smooth map between two regular parametric surfaces, let $\gamma : (t_0-\varepsilon, t_0+\varepsilon) \longrightarrow S_1,\ t \mapsto \gamma(t) \triangleq \mathbf{r}_1\big(u_1(t), v_1(t)\big)$ be a curve on S_1, which induces another curve $\varphi \circ \gamma$ on S_2, then we define the tangent map of φ at $p = \mathbf{r}_1\big(u(t_0), v(t_0)\big) \in S_1$ to be

$$\mathrm{d}\varphi_p : T_pS_1 \longrightarrow T_{\varphi(p)}S_2$$
$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=t_0} \gamma(t) \mapsto \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=t_0} (\varphi \circ \gamma)(t).$$

Proposition 1. For a smooth map $\varphi : S_1 \longrightarrow S_2$ between two regular parametric surfaces, its tangent map $\mathrm{d}\varphi_p : T_pS_1 \longrightarrow T_{\varphi(p)}S_2$ is a linear map such that

$$\mathrm{d}\varphi_p \begin{pmatrix} \partial_{u_1}\mathbf{r}_1 \\ \partial_{v_1}\mathbf{r}_1 \end{pmatrix} = \begin{pmatrix} \frac{\partial u_2}{\partial u_1} & \frac{\partial v_2}{\partial u_1} \\ \frac{\partial u_2}{\partial v_1} & \frac{\partial v_2}{\partial v_1} \end{pmatrix} \begin{pmatrix} \partial_{u_2}\mathbf{r}_2 \\ \partial_{v_2}\mathbf{r}_2 \end{pmatrix}.$$

Proof: By the discussion above, we see that

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}(\varphi\circ\gamma)(t) &= \frac{\partial \mathbf{r}_2}{\partial u_2}\cdot\frac{\mathrm{d}u_2(t)}{\mathrm{d}t}+\frac{\partial \mathbf{r}_2}{\partial v_2}\cdot\frac{\mathrm{d}v_2(t)}{\mathrm{d}t}\\ &= \frac{\partial \mathbf{r}_2}{\partial u_2}\cdot\left(\frac{\partial u_2}{\partial u_1}\cdot\frac{\mathrm{d}u_1(t)}{\mathrm{d}t}+\frac{\partial u_2}{\partial v_1}\cdot\frac{\mathrm{d}v_1(t)}{\mathrm{d}t}\right)+\frac{\partial \mathbf{r}_2}{\partial v_2}\cdot\left(\frac{\partial v_2}{\partial u_1}\cdot\frac{\mathrm{d}u_1(t)}{\mathrm{d}t}+\frac{\partial v_2}{\partial v_1}\cdot\frac{\mathrm{d}v_1(t)}{\mathrm{d}t}\right)\\ &= \left(\frac{\partial u_2}{\partial u_1}\cdot\partial_{u_2}\mathbf{r}_2+\frac{\partial v_2}{\partial u_1}\cdot\partial_{v_2}\mathbf{r}_2\right)\cdot\frac{\mathrm{d}u_1(t)}{\mathrm{d}t}+\left(\frac{\partial u_2}{\partial v_1}\cdot\partial_{u_2}\mathbf{r}_2+\frac{\partial v_2}{\partial v_1}\cdot\partial_{v_2}\mathbf{r}_2\right)\cdot\frac{\mathrm{d}v_1(t)}{\mathrm{d}t}.\end{aligned}$$

On the other hand,

$$\begin{aligned}\mathrm{d}\varphi_p\left(\frac{\mathrm{d}}{\mathrm{d}t}\gamma(t)\right) &= \mathrm{d}\varphi_p\left(\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{r}_1(u_1(t),v_1(t))\right)\\ &= \mathrm{d}\varphi_p\left(\partial_{u_1}\mathbf{r}_1\cdot\frac{\mathrm{d}u_1(t)}{\mathrm{d}t}+\partial_{v_1}\mathbf{r}_1\cdot\frac{\mathrm{d}v_1}{\mathrm{d}t}\right)\\ &= \mathrm{d}\varphi_p(\partial_{u_1}\mathbf{r}_1)\cdot\frac{\mathrm{d}u_1(t)}{\mathrm{d}t}+\mathrm{d}\varphi_p(\partial_{v_1}\mathbf{r}_1)\cdot\frac{\mathrm{d}v_1(t)}{\mathrm{d}t}.\end{aligned}$$

So by deninition $\mathrm{d}\varphi_p\left(\frac{\mathrm{d}}{\mathrm{d}t}\gamma(t)\right)=\frac{\mathrm{d}}{\mathrm{d}t}(\varphi\circ\gamma)(t)$ and compare these two equations, we get

$$\begin{cases}\mathrm{d}\varphi_p(\partial_{u_1}\mathbf{r}_1)=\frac{\partial u_2}{\partial u_1}\cdot\partial_{u_2}\mathbf{r}_2+\frac{\partial v_2}{\partial u_1}\cdot\partial_{v_2}\mathbf{r}_2\\ \mathrm{d}\varphi_p(\partial_{v_1}\mathbf{r}_1)=\frac{\partial u_2}{\partial v_1}\cdot\partial_{u_2}\mathbf{r}_2+\frac{\partial v_2}{\partial v_1}\cdot\partial_{v_2}\mathbf{r}_2\end{cases},$$

i.e., $\mathrm{d}\varphi_p\begin{pmatrix}\partial_{u_1}\mathbf{r}_1\\ \partial_{v_1}\mathbf{r}_1\end{pmatrix}=\begin{pmatrix}\frac{\partial u_2}{\partial u_1}&\frac{\partial v_2}{\partial u_1}\\ \frac{\partial u_2}{\partial v_1}&\frac{\partial v_2}{\partial v_1}\end{pmatrix}\begin{pmatrix}\partial_{u_2}\mathbf{r}_2\\ \partial_{v_2}\mathbf{r}_2\end{pmatrix}$. ■

Corollary 1.

The tangent map $\mathrm{d}\varphi_p : T_pS_1 \longrightarrow T_{\varphi(p)}S_2$ is an isomorphism $\Longleftrightarrow$ $\det\begin{pmatrix}\frac{\partial u_2}{\partial u_1}&\frac{\partial v_2}{\partial u_1}\\ \frac{\partial u_2}{\partial v_1}&\frac{\partial v_2}{\partial v_1}\end{pmatrix}=\frac{\partial(u_2,v_2)}{\partial(u_1,v_1)}\neq 0$.

Proof: ✓ ■

- Let ω be a quadratic differential form on a regular parametric surface S_2, which is written as

$$\omega \triangleq A(u_2,v_2)(\mathrm{d}u_2)^2+2B(u_2,v_2)\mathrm{d}u_2\mathrm{d}v_2+C(u_2,v_2)(\mathrm{d}v_2)^2,$$

then there is another quadratic differential form $\varphi^*\omega$, which is called the ***pullback of ω under φ***, on S_1 under the smooth map $\varphi : S_1 \longrightarrow S_2$, which is given by

$$\begin{aligned}\varphi^*\omega &= A\big(u_2(u_1,v_1),v_2(u_1,v_1)\big)\left(\frac{\partial u_2}{\partial u_1}\mathrm{d}u_1+\frac{\partial u_2}{\partial v_1}\mathrm{d}v_1\right)^2+\cdots\cdots\\ &\triangleq \widetilde{A}(u_1,v_1)(\mathrm{d}u_1)^2+2\widetilde{B}(u_1,v_1)\mathrm{d}u_1\mathrm{d}v_1+\widetilde{C}(u_1,v_1)(\mathrm{d}v_1)^2.\end{aligned}$$

By some computation, we can get

$$\begin{pmatrix}\widetilde{A}(u_1,v_1)&\widetilde{B}(u_1,v_1)\\ \widetilde{B}(u_1,v_1)&\widetilde{C}(u_1,v_1)\end{pmatrix}=\begin{pmatrix}\frac{\partial u_2}{\partial u_1}&\frac{\partial v_2}{\partial u_1}\\ \frac{\partial u_2}{\partial v_1}&\frac{\partial v_2}{\partial v_1}\end{pmatrix}\begin{pmatrix}A\big(u_2(u_1,v_1),v_2(u_1,v_1)\big)&B\big(u_2(u_1,v_1),v_2(u_1,v_1)\big)\\ B\big(u_2(u_1,v_1),v_2(u_1,v_1)\big)&C\big(u_2(u_1,v_1),v_2(u_1,v_1)\big)\end{pmatrix}\begin{pmatrix}\frac{\partial u_2}{\partial u_1}&\frac{\partial v_2}{\partial u_1}\\ \frac{\partial u_2}{\partial v_1}&\frac{\partial v_2}{\partial v_1}\end{pmatrix}^{\mathrm{T}},$$

we denote the matrix $J:=\begin{pmatrix}\frac{\partial u_2}{\partial u_1}&\frac{\partial v_2}{\partial u_1}\\ \frac{\partial u_2}{\partial v_1}&\frac{\partial v_2}{\partial v_1}\end{pmatrix}$, hence we have

$$\begin{pmatrix}\widetilde{A}(u_1,v_1)&\widetilde{B}(u_1,v_1)\\ \widetilde{B}(u_1,v_1)&\widetilde{C}(u_1,v_1)\end{pmatrix}=J\begin{pmatrix}A\big(u_2(u_1,v_1),v_2(u_1,v_1)\big)&B\big(u_2(u_1,v_1),v_2(u_1,v_1)\big)\\ B\big(u_2(u_1,v_1),v_2(u_1,v_1)\big)&C\big(u_2(u_1,v_1),v_2(u_1,v_1)\big)\end{pmatrix}J^{\mathrm{T}}.$$

By these computation, we immediately get the following consequence.

Proposition 2. Let $\varphi : S_1 \longrightarrow S_2$, $(u_1, v_1) \mapsto (u_2, v_2) = \big(u_2(u_1, v_1), v_2(u_1, v_1)\big)$ be a smooth map between two regular parametric surfaces, for a quadratic differential form ω on S_2, if

$$\omega \triangleq \begin{pmatrix} \mathrm{d}u_2 & \mathrm{d}v_2 \end{pmatrix} \begin{pmatrix} A(u_2, v_2) & B(u_2, v_2) \\ B(u_2, v_2) & C(u_2, v_2) \end{pmatrix} \begin{pmatrix} \mathrm{d}u_2 \\ \mathrm{d}v_2 \end{pmatrix},$$

then the pullback $\varphi^*\omega$ on S_1 is

$$\varphi^*\omega = \begin{pmatrix} \mathrm{d}u_1 & \mathrm{d}v_1 \end{pmatrix} J \begin{pmatrix} A\big(u_2(u_1, v_1), v_2(u_1, v_1)\big) & B\big(u_2(u_1, v_1), v_2(u_1, v_1)\big) \\ B\big(u_2(u_1, v_1), v_2(u_1, v_1)\big) & C\big(u_2(u_1, v_1), v_2(u_1, v_1)\big) \end{pmatrix} J^{\mathrm{T}} \begin{pmatrix} \mathrm{d}u_1 \\ \mathrm{d}v_1 \end{pmatrix},$$

where we denote $J := \begin{pmatrix} \frac{\partial u_2}{\partial u_1} & \frac{\partial v_2}{\partial u_1} \\ \frac{\partial u_2}{\partial v_1} & \frac{\partial v_2}{\partial v_1} \end{pmatrix}$.

Definition 2.(Isometry between Regular Parametrized Surfaces)
Let $\varphi : S_1 \longrightarrow S_2$ be a smooth map between regular parametric surfaces, if for $\forall p \in S_1$, the tangent map $\mathrm{d}\varphi_p : T_pS_1 \longrightarrow T_{\varphi(p)}S_2$ is length-preserving, i.e.,

$$\|\mathrm{d}\varphi_p(v)\| = \|v\| \quad \text{for } \forall v \in T_pS_1,$$

then we say φ is an ***isometry*** (or ***length-preserving correspondence***).

Remark: If $\varphi : S_1 \longrightarrow S_2$ is an isometry, then the tangent map $\mathrm{d}\varphi_p : T_pS_1 \longrightarrow T_{\varphi(p)}S_2$ also preserves the inner products of any two tangent vectors, i.e.,

$$\mathrm{d}\varphi_p(v_1) \cdot \mathrm{d}\varphi_p(v_2) = v_1 \cdot v_2 \quad \text{for } \forall v_1, v_2 \in T_pS_1.$$

Proof: Only need to note that

$$v_1 \cdot v_2 = \frac{1}{2}\left(\|v_1 + v_2\|^2 - \|v_1\|^2 - \|v_2\|^2\right),$$

hence

$$\begin{aligned} \mathrm{d}\varphi_p(v_1) \cdot \mathrm{d}\varphi_p(v_2) &= \frac{1}{2}\left(\|\mathrm{d}\varphi_p(v_1) + \mathrm{d}\varphi_p(v_2)\|^2 - \|\mathrm{d}\varphi_p(v_1)\|^2 - \|\mathrm{d}\varphi_p(v_2)\|^2\right) \\ &= \frac{1}{2}\left(\|\mathrm{d}\varphi_p(v_1 + v_2)\|^2 - \|\mathrm{d}\varphi_p(v_1)\|^2 - \|\mathrm{d}\varphi_p(v_2)\|^2\right) \\ &= \frac{1}{2}\left(\|v_1 + v_2\|^2 - \|v_1\|^2 - \|v_2\|^2\right) \\ &= v_1 \cdot v_2. \end{aligned}$$

which completes the proof. ■

Theorem 1. Let $\varphi : S_1 \longrightarrow S_2$ be a smooth map between two regular parametric surfaces, then

$$\varphi \text{ is an isometry } \iff \varphi^*\mathrm{I}_2 = \mathrm{I}_1,$$

where I_1 and I_2 are the first fundamental forms on S_1 and S_2 respectively.

Proof: Denote $S_1 : \mathbf{r}_1 = \mathbf{r}_1(u_1, v_1)$, $S_2 : \mathbf{r}_2 = \mathbf{r}_2(u_2, v_2)$, then by definition we get

$$\mathrm{I}_1 = \mathrm{d}\mathbf{r}_1(u_1, v_1) \cdot \mathrm{d}\mathbf{r}_1(u_1, v_1) = \|\mathrm{d}\mathbf{r}_1(u_1, v_1)\|^2 = \|\mathrm{d}\varphi_p(\mathrm{d}\mathbf{r}_1(u_1, v_1))\|^2,$$

where we denote $p \triangleq \mathbf{r}_1(u_1, v_1) \in S_1$. Since $(\mathrm{d}u_1, \mathrm{d}v_1)$ is the components of the tangent vector $\mathrm{d}\mathbf{r}_1(u_1, v_1)$ under the basis $\{\partial_{u_1}\mathbf{r}_1, \partial_{v_1}\mathbf{r}_1\}$, we have

$$\begin{aligned}
\mathrm{d}\varphi_p\big(\mathrm{d}\mathbf{r}_1(u_1, v_1)\big) &= \mathrm{d}\varphi_p\left(\begin{pmatrix}\mathrm{d}u_1 & \mathrm{d}v_1\end{pmatrix}\begin{pmatrix}\partial_{u_1}\mathbf{r}_1 \\ \partial_{v_1}\mathbf{r}_1\end{pmatrix}\right) \\
&= \begin{pmatrix}\mathrm{d}u_1 & \mathrm{d}v_1\end{pmatrix}\mathrm{d}\varphi_p\begin{pmatrix}\partial_{u_1}\mathbf{r}_1 \\ \partial_{v_1}\mathbf{r}_1\end{pmatrix} \\
&= \begin{pmatrix}\mathrm{d}u_1 & \mathrm{d}v_1\end{pmatrix} J \begin{pmatrix}\partial_{u_2}\mathbf{r}_2 \\ \partial_{v_2}\mathbf{r}_2\end{pmatrix}, \quad \text{(by Proposition 1)}
\end{aligned}$$

where $J := \begin{pmatrix}\frac{\partial u_2}{\partial u_1} & \frac{\partial v_2}{\partial u_1} \\ \frac{\partial u_2}{\partial v_1} & \frac{\partial v_2}{\partial v_1}\end{pmatrix}$. Thus we see that

$$\begin{aligned}
\mathrm{I}_1 = \big\|\mathrm{d}\varphi_p\big(\mathrm{d}\mathbf{r}_1(u_1, v_1)\big)\big\|^2 &= \begin{pmatrix}\mathrm{d}u_1 & \mathrm{d}v_1\end{pmatrix} J \begin{pmatrix}\partial_{u_2}\mathbf{r}_2 \\ \partial_{v_2}\mathbf{r}_2\end{pmatrix} \cdot \left(\begin{pmatrix}\mathrm{d}u_1 & \mathrm{d}v_1\end{pmatrix} J \begin{pmatrix}\partial_{u_2}\mathbf{r}_2 \\ \partial_{v_2}\mathbf{r}_2\end{pmatrix}\right)^{\mathrm{T}} \\
&= \begin{pmatrix}\mathrm{d}u_1 & \mathrm{d}v_1\end{pmatrix} J \begin{pmatrix}\partial_{u_2}\mathbf{r}_2 \\ \partial_{v_2}\mathbf{r}_2\end{pmatrix}\begin{pmatrix}\partial_{u_2}\mathbf{r}_2 & \partial_{v_2}\mathbf{r}_2\end{pmatrix} J^{\mathrm{T}}\begin{pmatrix}\mathrm{d}u_1 \\ \mathrm{d}v_1\end{pmatrix} \\
&= \begin{pmatrix}\mathrm{d}u_1 & \mathrm{d}v_1\end{pmatrix} J \begin{pmatrix}E_2(u_2, v_2) & F_2(u_2, v_2) \\ F_2(u_2, v_2) & G_2(u_2, v_2)\end{pmatrix} J^{\mathrm{T}}\begin{pmatrix}\mathrm{d}u_1 \\ \mathrm{d}v_1\end{pmatrix} \\
&= \varphi^*\mathrm{I}_2. \quad (\text{ by Proposition 2})
\end{aligned}$$

This completes the proof. ■

Corollary 2. Let $\varphi : S_1 \longrightarrow S_2$ be a smooth map between two regular parametric surfaces, then we have

$$\begin{pmatrix}E_1(u_1, v_1) & F_1(u_1, v_1) \\ F_1(u_1, v_1) & G_1(u_1, v_1)\end{pmatrix} = J\begin{pmatrix}E_2(u_2, v_2) & F_2(u_2, v_2) \\ F_2(u_2, v_2) & G_2(u_2, v_2)\end{pmatrix} J^{\mathrm{T}},$$

where $J := \begin{pmatrix}\frac{\partial u_2}{\partial u_1} & \frac{\partial v_2}{\partial u_1} \\ \frac{\partial u_2}{\partial v_1} & \frac{\partial v_2}{\partial v_1}\end{pmatrix}$.

Proof: ✓ ■

Lemma 1. Let $\varphi : S_1 \longrightarrow S_2$ be a smooth map between regular parametric surfaces, $p \in S_1$. If the tanggent map $\mathrm{d}\varphi_p : T_pS_1 \longrightarrow T_{\varphi(p)}S_2$ is an isomorphism, then there exists a neighborhood U_1 of p on S_1 and a neighborhood U_2 of $\varphi(p)$ on S_2 such that

$$\varphi(U_1) \subseteq U_2 \text{ and } \begin{cases} u_2(u_1, v_1) = u_1, \\ v_2(u_1, v_1) = v_1. \end{cases}$$

Proof: Since $\mathrm{d}\varphi_p$ is a linear isomorphism, hence we have

$$\det\begin{pmatrix} \frac{\partial u_2}{\partial u_1} & \frac{\partial v_2}{\partial u_1} \\ \frac{\partial u_2}{\partial v_1} & \frac{\partial v_2}{\partial v_1} \end{pmatrix} \neq 0,$$

hence $\begin{cases} u_2 = u_2(u_1, v_1) \\ v_2 = v_2(u_1, , v_2) \end{cases}$ can be viewed as an admissible change of parameters, which will turn (u_2, v_2) be a new local coordinate of p on S_1, in which case $\varphi|_{U_1} = \mathrm{Id}$ and $J = I_2$. ■

Remark: Such parametrizations $(u_1, v_1), (u_2, v_2)$ as in the lemma above are called ***adapted parameter system*** of S_1 and S_2 under φ.

Corollary 3. Let S_1, S_2 be two regular parametric surfaces, then there is an isometry $\varphi : S_1 \longrightarrow S_2 \Longleftrightarrow$ there is an adapted parameter system (u, v) for both S_1 and S_2 such that $\begin{cases} E_1(u,v) = E_2(u,v), \\ F_1(u,v) = F_2(u,v), \\ G_1(u,v) = G_2(u,v). \end{cases}$ i.e., the first fundamental coefficients of S_1 and S_2 are the same.

Proof: ✓ ■